THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · MEDIUM LEVEL UNIT TEST SET 1

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. In the pattern $4, 11, 18, 25, \ldots$ each term is 7 more than the one before. The 12th term is: 1

(a) 84 **(b)** 88 **(c)** 81 **(d)** 95

2. What is the angle between the hands of a clock at 4 o'clock, measured the long way round (the larger angle)? 1

(a) 240 **(b)** 270 **(c)** 210 **(d)** 120

3. The smallest EVEN number that can be formed by using each of the digits 3, 1, 4, 2 once is: 1

(a) 1342 **(b)** 1324 **(c)** 4312 **(d)** 1234

4. The total of the two largest values in the table is: 1

Item	Number
Apple	10
Banana	70
Orange	30

(a) 40 **(b)** 80 **(c)** 100 **(d)** 105

5. The number of factors of 48 is: 1

(a) 4 **(b)** 5 **(c)** 10 **(d)** 12

6. In the pattern $3, 8, 13, 18, \ldots$ which term number equals 88? 1

(a) 17 **(b)** 23 **(c)** 18 **(d)** 19

7. The minute hand starts at 12 and stops at 4. The angle it turns through (clockwise) is: 1

(a) 90 **(b)** 240 **(c)** 150 **(d)** 120

8. The number of 3-digit palindromes is: 1

(a) 100 **(b)** 99 **(c)** 90 **(d)** 81

9. A pictograph has one symbol = 15. The number of symbols for A is 9 and for B is 4. How many more does A have than B? 1

(a) 5 **(b)** 90 **(c)** 75 **(d)** 195

10. The HCF of 36 and 48 is: 1

(a) 36 **(b)** 12 **(c)** 144 **(d)** 24

11. What is the sum of the first 12 even numbers $2 + 4 + 6 + \ldots$? 1

(a) 78 **(b)** 132 **(c)** 144 **(d)** 156

12. The hour hand of a clock turns through 150°. How many hours have passed? 1

(a) 7 **(b)** 5 **(c)** 2 **(d)** 10

13. The greatest 5-digit palindrome with first digit 2 is: 1

(a) 99999 **(b)** 29892 **(c)** 2992 **(d)** 29992

14. A bar of height 6 cm in a graph with scale 1 cm = 15 units, and another bar of 5 cm. Together they show: 1

(a) 26 **(b)** 165 **(c)** 180 **(d)** 150

15. The LCM of 12 and 18 is: 1

(a) 216 **(b)** 72 **(c)** 18 **(d)** 36

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. In the pattern $5, 8, 11, 14, \ldots$ state the rule for getting any term from the first term, and find the 20th term. 2

17. Find the smaller and the larger angle between the hands of a clock at 5 o'clock, and name their types. 2

18. Find the greatest and the smallest 5-digit numbers formed from 4, 9, 1, 7, 3, each digit once. 2

19. Find the difference between the highest and lowest and the total of all. 2

Item	Number
Apple	75
Mango	70
Orange	35

20. Find the HCF and LCM of 24 and 36 using prime factorisation. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. A row of triangles is made with sticks: 3 sticks for the first triangle and 2 more for each new triangle. Find the sticks needed for 12 triangles, and the number of triangles that can be made with 35 sticks. 3

22. A ray OA is turned about O through a three-quarter turn clockwise and then through a quarter turn anticlockwise. Find the net turn in degrees and its type. 3

23. Perform two Kaprekar steps on 1795. 3

24. Find the fraction of students liking Kabaddi and the number who do not like it. 3

Sport	Students
Kabaddi	25
Chess	40
Tennis	65

OR

Find the prime factorisation of 180 and the number of its factors. 3

25. Find the sum of all odd numbers from 1 to 59. Hence find the sum of all even numbers from 2 to 60. 3

OR

A ray OA is turned about O: first 60° anticlockwise, then 70° anticlockwise, then 40° clockwise. Find the net angle of turn from the original position. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. A pattern of squares: the first figure has 5 squares and each next figure has 4 more. Find the number of squares in the 10th figure, and the first figure that has more than 91 squares. 5

OR

A clock shows 1:00. The hour hand turns clockwise until the clock shows 3:00. Find the angle turned by the hour hand, the angle turned by the minute hand in this time, and the smaller angle between the hands at 3:00. 5

27. A 4-digit palindrome has first digit 2 and digit sum 10. Find the number. Add 11 to it and see if the answer is a palindrome. 5

OR

A pictograph (one symbol = 12) shows 3 symbols on Monday, 6 on Tuesday and 9 on Wednesday. On Thursday the number of items sold equals the total of Monday and Tuesday. Find the number sold on each day, and the total for the four days. 5

END OF QUESTION PAPER Total: 50 Marks

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